TTC Delay Data pattern Analysis

1. Project Overview

This project analyzes TTC delay behavior using delay data from official TTC website for the year 2025. The goal is to uncover the reasons that causes the most delays across all stations.

2. Dataset Summary

Rows: 27,713 delay incidents

Columns: 14

Data Types:

- 1 Integer columns
- 1 Float column
- 9 Categorical/Text columns

Time Period: 2025

Granularity: Incident-level operational event data

3. Key Features: -

id_x – Unique incident record ID

id_y – Foreign key linking to equipment fault dataset

Code / CODE – Operational delay code

DESCRIPTION – Detailed explanation of delay cause

Station – Station where incident occurred

Line – Subway line (e.g., BD, YU)

Bound – Direction of travel (E, W, etc.)

Min Delay – Minutes of service delay

Min Gap – Minutes between vehicles due to disruption

Vehicle – Vehicle number involved

4. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used **df.info ()** to check structure and **.describe()** for summary statistics.

```
df.describe(include='all')
```

	_id_x	Date	Time	Day	Station	Code	Min Delay	Min Gap	Bound	Line	Vehi
count	25713.000000	25713	25713	25713	25713	25713	25713.000000	25713.000000	16251	25646	25713.0000
unique	NaN	365	1426	7	475	131	NaN	NaN	5	18	N
top	NaN	2025-07-30	22:00	Wednesday	KENNEDY BD STATION	SUDP	NaN	NaN	S	YU	N
freq	NaN	144	373	4035	991	2945	NaN	NaN	4357	13079	N
mean	12857.000000	NaN	NaN	NaN	NaN	NaN	2.751682	3.908645	NaN	NaN	3201.1752
std	7422.848072	NaN	NaN	NaN	NaN	NaN	11.101464	11.742261	NaN	NaN	2736.5439
min	1.000000	NaN	NaN	NaN	NaN	NaN	0.000000	0.000000	NaN	NaN	0.0000
25%	6429.000000	NaN	NaN	NaN	NaN	NaN	0.000000	0.000000	NaN	NaN	0.0000
50%	12857.000000	NaN	NaN	NaN	NaN	NaN	0.000000	0.000000	NaN	NaN	5110.0000
75%	19285.000000	NaN	NaN	NaN	NaN	NaN	4.000000	7.000000	NaN	NaN	5551.0000
max	25713.000000	NaN	NaN	NaN	NaN	NaN	900.000000	906.000000	NaN	NaN	9463.0000

_id_y	CODE	DESCRIPTION
5581.000000	25581	25581
NaN	128	128
NaN	SUDP	DISORDERLY PATRON
NaN	2945	2945
78.693405	NaN	NaN
35.612582	NaN	NaN
1.000000	NaN	NaN
47.000000	NaN	NaN
73.000000	NaN	NaN
114.000000	NaN	NaN
140.000000	NaN	NaN

Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- Column Standardization: Renamed columns to snake case for better readability and documentation.
- Feature Engineering: Created categories like UNKNOWN column by binning line and CODE.
- Database Integration: Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

5. Data Analysis using SQL (Detailed Analysis)

1. Reason that causes most delays

code	description	incident_count	total_delay_minutes	avg_delay
SUDP	DISORDERLY PATRON	2945	7658	2.60
MUIR	INJURED/ILL CUSTOMER ON TRAIN - MEDICAL ...	842	4807	5.71
SUUT	UNAUTHORIZED AT TRACK LEVEL	571	3917	6.86
MUI	INJURED/ILL CUSTOMER ON TRAIN - TRANSPO...	411	3785	9.21
PUOPO	OPTO (COMMUNICATIONS) TRAIN DOOR MONI...	1264	3426	2.71
SUO	SECURITY OTHER	1672	3418	2.04

2. Stations which are risk zone areas

station	incident_count	total_delay	avg_delay
EGLINTON STATION	650	2764	4.25
KIPLING STATION	864	2255	2.61
SHEPPARD WEST STATION	338	2214	6.55
KENNEDY BD STATION	982	2195	2.24
WILSON STATION	629	2163	3.44
FINCH STATION	898	1826	2.03

3. All Lines detailed analysis

line	incidents	total_delay	avg_delay
YU	12989	39572	3.05
SHP	1064	3107	2.92
BD	11104	27149	2.44
Unknown	65	3	0.05
YU/BD	319	0	0.00
YU/ BD	3	0	0.00
YU / BD	2	0	0.00
YUS	4	0	0.00
YUS/BD	15	0	0.00

4. Those delays which are severe with their root causes

code	incidents	severe_cases	severe_percentage
MUPR1	21	15	71.43
PUTTP	6	4	66.67
TUDOE	4	2	50.00
MUEC	2	1	50.00
MUPLB	46	18	39.13
PUTCD	3	1	33.33
PUSSW	60	17	28.33
PUSZC	4	1	25.00
PUSCA	4	1	25.00

5. Ascending to Descending order or delays according to station

code	total_delay	delay_rank
SUDP	7658	1
MUIR	4807	2
SUUT	3917	3
MUI	3785	4
PUOPO	3426	5
SUO	3418	6
MUPAA	2955	7
MUWEA	2745	8
PUTWZ	2341	9
TUO	1935	10
SUAP	1759	11

6. Causes that contribute 80% of delay incidents

code	total_delay	cumulative_pct
SUDP	7658	0.1097
MUIR	4807	0.1785
SUUT	3917	0.2346
MUI	3785	0.2888
PUOPO	3426	0.3379
SUO	3418	0.3868
MUPAA	2955	0.4291
MUWEA	2745	0.4684
PUTWZ	2341	0.5020

7. Stations that have highest delays

station	incidents	severe_cases	severe_rate
NORTH YORK CENTRE STAT	21	1	4.76
LAWRENCE STATION	313	8	2.56
DOWNSVIEW PARK STATION	153	3	1.96
SHEPPARD-YONGE STATION	310	6	1.94
SHEPPARD WEST STATION	338	6	1.78
YORK MILLS STATION	292	5	1.71
GREENWOOD STATION	301	5	1.66
LAWRENCE WEST STATION	251	4	1.59
ISLINGTON STATION	392	6	1.53
MUSEUM STATION	159	2	1.26

8. Does service gap severity predict delay severity?

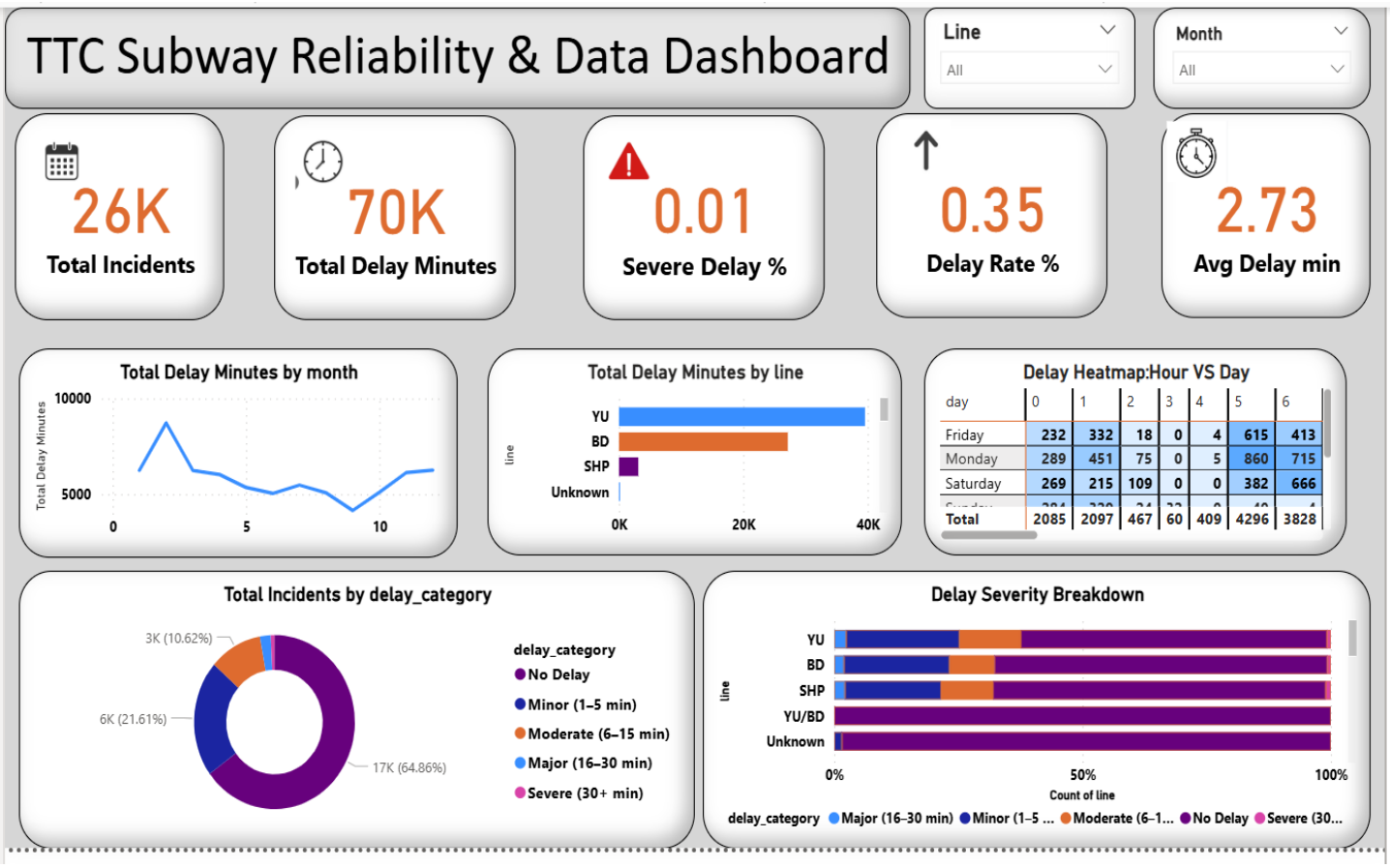
gap_category	avg_delay	severe_rate
Severe Gap (30+ min)	62.24	70.26
Major Gap (16–30 min)	15.75	0.00
Moderate Gap (6–15 min)	4.93	0.00
Minor Gap (1–5 min)	3.01	0.00
No Gap Issue	0.30	0.14

9. What % of total delay is caused by top 5 stations?

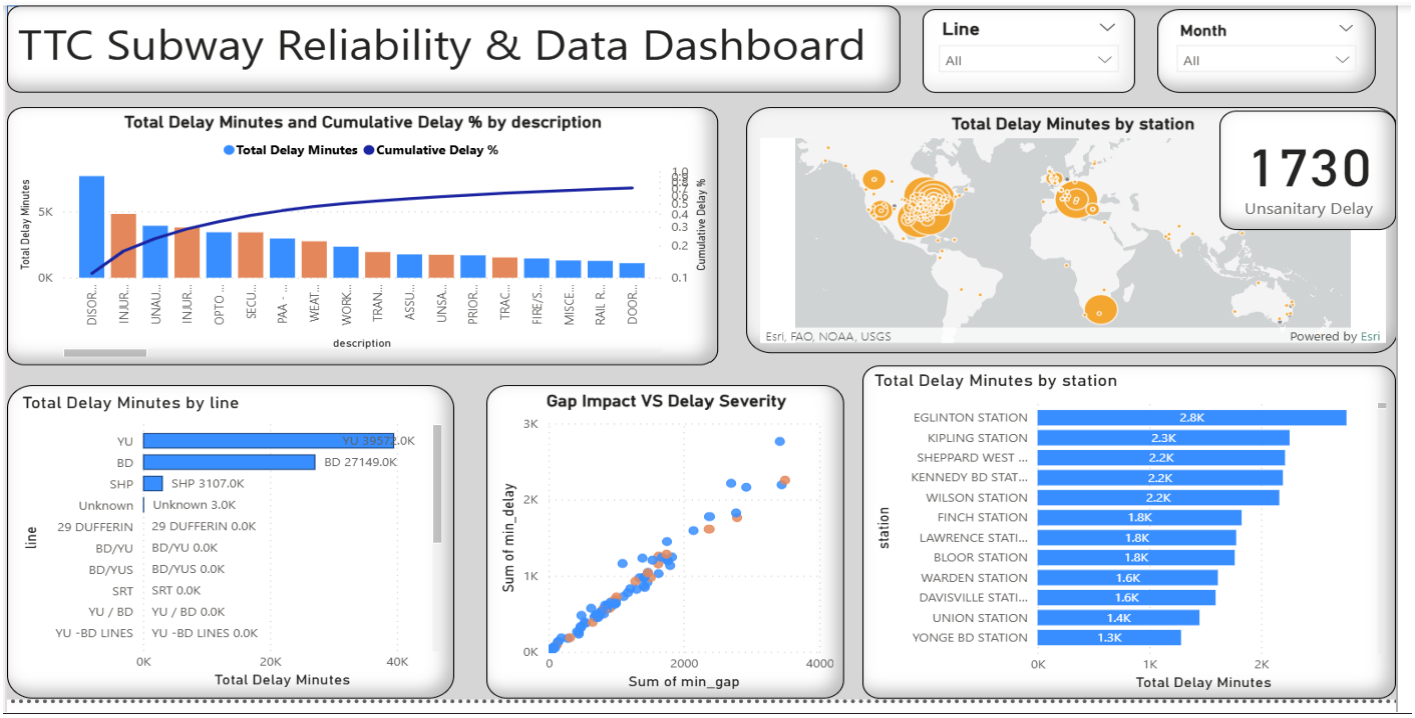
top5_contribution
16.5986

6. Dashboard in Power BI

Page 1: Executive Overview



Page 2: Root Cause Analysis



7. Business Recommendations

- **Prioritize High-Impact Lines:** The Yonge-University line contributes the largest share of total delay minutes. Operational audits and targeted reliability initiatives should focus on this corridor to maximize network-wide impact.

- **Address Top Delay Drivers (Pareto Insight):** A small number of delay descriptions contribute to a majority of cumulative delay minutes. Implement root-cause mitigation programs targeting the top 20% delay categories to reduce overall delay burden efficiently.

- **Improve Preventive Maintenance Strategy:** Frequent minor delays (1–5 minutes) dominate incident distribution. Strengthening preventive maintenance schedules could reduce recurring minor disruptions and improve overall service reliability.

- **Reduce Severe Delay Frequency:** Although severe delays (>30 min) represent a smaller proportion of total incidents, they significantly impact passenger satisfaction. Introduce rapid-response protocols and contingency routing plans for severe disruptions.

- **Optimize Gap Management:** Strong correlation between service gap (min_gap) and total delay minutes suggests schedule adherence issues. Implement tighter headway monitoring and real-time operational control mechanisms.